

卡拉 OK 机设计

1、实验目的：

使用 HX4S20 开发板上接扩展板麦克风输入、线路输入以及线路输出端口组成一个卡拉 OK 机。

2、实验原理：

Wolfson WM8731 音频编解码芯片配置为主机模式，在这种模式下，编解码芯片自己生成 AD/DA 串行位时钟（BCK）以及左/右声道时钟信号（LRCK）。正如图 2.1 所示，芯片的 I2C 界面用于配置音频 CODEC。CODEC 芯片的采样率就是通过这个接口来设定的，从线路输入的音频信号会和从麦克风输入的信号混合，然后送到线路输出端口。

在这个设计范例中，采样率设置为 48kHz。按下 KEY0 会通过 I2C 总线调整音频编解码芯片的增益，增益值在 10 个预定义的值间循环。

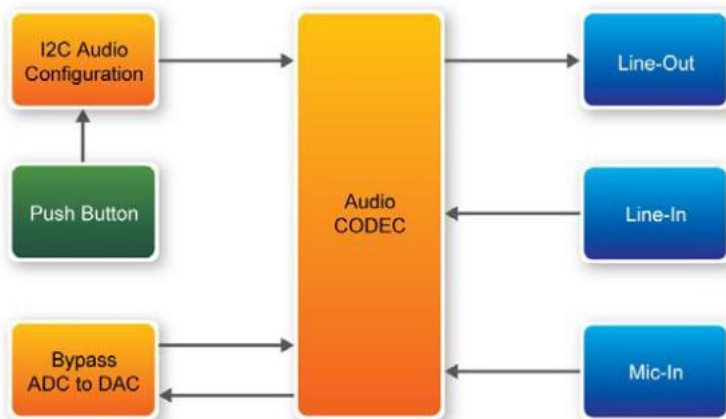


图 2.1

3、实验程序：

```
module i2c (
    CLOCK,
    I2C_SCLK,    //I2C CLOCK
    I2C_SDAT,    //I2C DATA
    I2C_DATA,    //DATA: [SLAVE_ADDR, SUB_ADDR, DATA]
    GO,          //GO transfor
    END,         //END transfor
    W_R,         //W_R
    ACK,         //ACK
```

```

        RESET,
        //TEST
        SD_COUNTER,
        SDO
    );

//=====
//  PORT declarations
//=====

    input          CLOCK;
    input          [23:0] I2C_DATA;
    input          GO;
    input          RESET;
    input          W_R;

    inout          I2C_SDAT;

    output         I2C_SCLK;
    output         END;
    output         ACK;

//TEST
    output [5:0]   SD_COUNTER;
    output         SDO;

    reg            SDO;
    reg            SCLK;
    reg            END;
    reg [23:0]     SD;
    reg [5:0]      SD_COUNTER;

wire I2C_SCLK = SCLK | ( ((SD_COUNTER >= 4) & (SD_COUNTER <= 30)) ? ~CLOCK : 0 );
wire I2C_SDAT = SDO?1'bz:0 ;

reg ACK1,ACK2,ACK3;
wire ACK = ACK1 | ACK2 | ACK3;

//=====
//  Structural coding
//=====

//=====I2C COUNTER=====

```

```

always @(negedge RESET or posedge CLOCK )
begin
    if (!RESET)
        begin
            SD_COUNTER = 6'b111111;
        end
    else begin
        if (GO == 0)
            begin
                SD_COUNTER = 0;
            end
        else begin
            if (SD_COUNTER < 6'b111111)
                begin
                    SD_COUNTER = SD_COUNTER+1;
                end
            end
        end
    end
end

//=====I2C COUNTER=====

always @(negedge RESET or posedge CLOCK )
begin
    if (!RESET)
        begin
            SCLK = 1;
            SDO = 1;
            ACK1 = 0;
            ACK2 = 0;
            ACK3 = 0;
            END = 1;
        end
    else
        case (SD_COUNTER)
            6'd0 : begin
                ACK1 = 0 ;
                ACK2 = 0 ;
                ACK3 = 0 ;
                END = 0 ;
                SDO =1 ;
                SCLK =1 ;
            end
            //=====start=====
            6'd1 : begin

```

```

        SD = I2C_DATA;
        SDO = 0;
    end

6' d2 :    SCLK = 0;
//=====SLAVE ADDR=====
6' d3 :    SDO = SD[23];
6' d4 :    SDO = SD[22];
6' d5 :    SDO = SD[21];
6' d6 :    SDO = SD[20];
6' d7 :    SDO = SD[19];
6' d8 :    SDO = SD[18];
6' d9 :    SDO = SD[17];
6' d10 :   SDO = SD[16];
6' d11 :   SDO = 1'b1;//ACK

//=====SUB ADDR=====
6' d12 : begin
            SDO = SD[15];
            ACK1 = I2C_SDAT;
        end
6' d13 :   SDO = SD[14];
6' d14 :   SDO = SD[13];
6' d15 :   SDO = SD[12];
6' d16 :   SDO = SD[11];
6' d17 :   SDO = SD[10];
6' d18 :   SDO = SD[9];
6' d19 :   SDO = SD[8];
6' d20 :   SDO = 1'b1;//ACK

//=====DATA=====
6' d21 : begin
            SDO = SD[7];
            ACK2 = I2C_SDAT;
        end
6' d22 :   SDO = SD[6];
6' d23 :   SDO = SD[5];
6' d24 :   SDO = SD[4];
6' d25 :   SDO = SD[3];
6' d26 :   SDO = SD[2];
6' d27 :   SDO = SD[1];
6' d28 :   SDO = SD[0];
6' d29 :   SDO = 1'b1;//ACK

```

```

        //stop
        6'd30 : begin
            SDO = 1'b0;
            SCLK = 1'b0;
            ACK3 = I2C_SDAT;
        end
        6'd31 : SCLK = 1'b1;
        6'd32 : begin
            SDO = 1'b1;
            END = 1;
        end

    endcase

end

endmodule

```

4 、实验仿真:

根据描述语言进行仿真测试，分析波形。

5 、管脚约束:

```

#IO 管脚约束
set_pin_assignment { aud_adcdat } { LOCATION = H16; IOSTANDARD = LVCMOS33; PULLTYPE
= PULLUP; }
set_pin_assignment { aud_adclrck } { LOCATION = M16; IOSTANDARD = LVCMOS33; PULLTYPE
= PULLUP; }
set_pin_assignment { aud_bclk } { LOCATION = F15; IOSTANDARD = LVCMOS33; PULLTYPE
= PULLUP; }
set_pin_assignment { aud_dacdat } { LOCATION = G14; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { aud_daclrck } { LOCATION = J16; IOSTANDARD = LVCMOS33; PULLTYPE
= PULLUP; }
set_pin_assignment { aud_xck } { LOCATION = E15; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { clock_50 } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE
= PULLUP; }
set_pin_assignment { i2c_dat } { LOCATION = P11; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = PULLUP; }
set_pin_assignment { i2c_sclk } { LOCATION = M10; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }

```

```

set_pin_assignment { key[0] } { LOCATION = B2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[1] } { LOCATION = A2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { keyon } { LOCATION = A4; IOSTANDARD = LVCMOS33; DRIVESTRENGTH
= 8; PULLTYPE = NONE; }

```

6 、下载测试

将语音模块接插在开发板 J1 口上，连接麦克风到开发板麦克风（MIC）输入端口（绿色插孔）。连接音乐播放器，譬如 MP3 播放器或者电脑的音频输入端口（粉红色插孔）。连接耳机或者有源音箱到开发板的线路（LINE OUT）输出端口（蓝色插孔）。

下载程序之后。可以听到耳机/音箱播放的来自麦克风和音乐播放器的混合声音。KEY1 复位， 按下 KEY2 调整音量， 音量大小会在 0-9 级间循环。**注意音量防止震伤耳朵。**

